

The Executive Overview

High-level view of fraud exposure, tracking KPIs, impact and infection rates.

KEY INSIGHT

Online shopping and grocery transactions carry the highest fraud rates at **1.44%** and **1.32%** - nearly 3× the overall average of **0.52%**. These two categories alone account for a disproportionate share of total fraud exposure and should be **the first targets for enhanced real-time verification**.

Clear all slicers

Category
All

Age Group
All

State
All

Transaction Type

Fraudulent

Legitimate

Amount Band
All

Total Transactions

340K

Total Fraud Cases

1.78K

Overall Fraud rate

0.52%

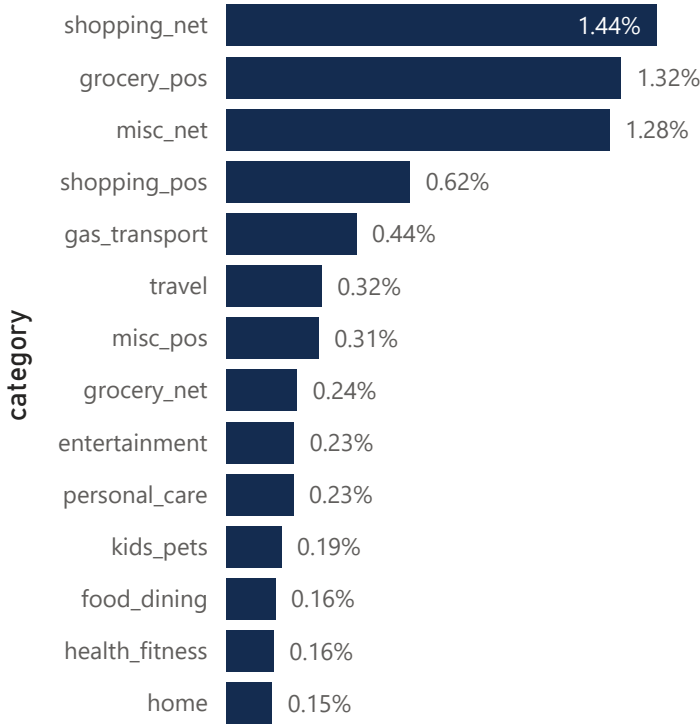
Average Fraud Amount

\$518.07

Date Range

1/1/2019 12/31/2020

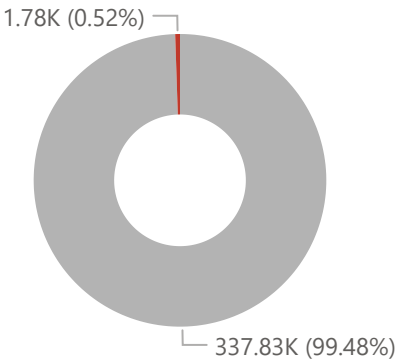
Fraud Rate by category



Total Fraud by State

Fraud and Legitimate Transactions

● Legitimate ● Fraudulent



Transactional Intelligence & Deep Dive

Analyzes when and how fraud occurs, revealing tactics via transaction size, distance, and timing patterns.

Executive Overview

Deep Dive

Risk Profile

Model Performance

Guide

KEY INSIGHT

Fraud concentrates sharply between **midnight and 3 AM**, with 11PM peaking at **2.62%** — five times the daytime average. Transactions **above \$500** carry a **20.70% fraud rate**, making high-value nighttime purchases the highest combined risk profile in this dataset.

Clear all slicers

Category

All

Age Group

All

State

All

Transaction Type

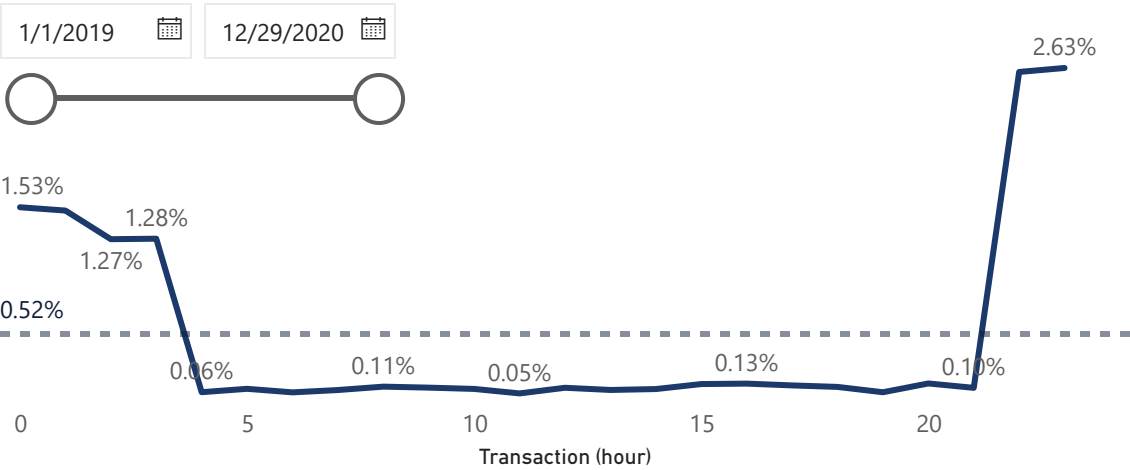
Fraudulent

Legitimate

Amount Band

All

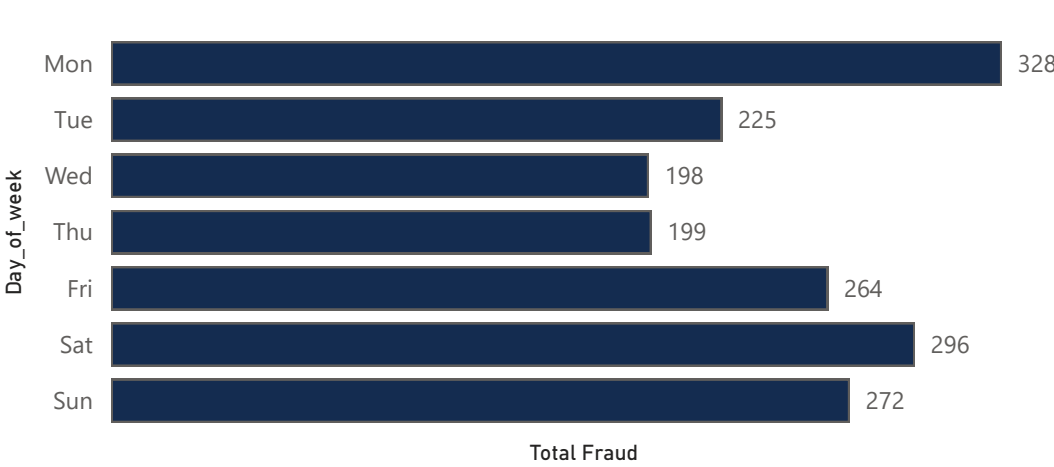
Fraud Rate by Transaction (hour)



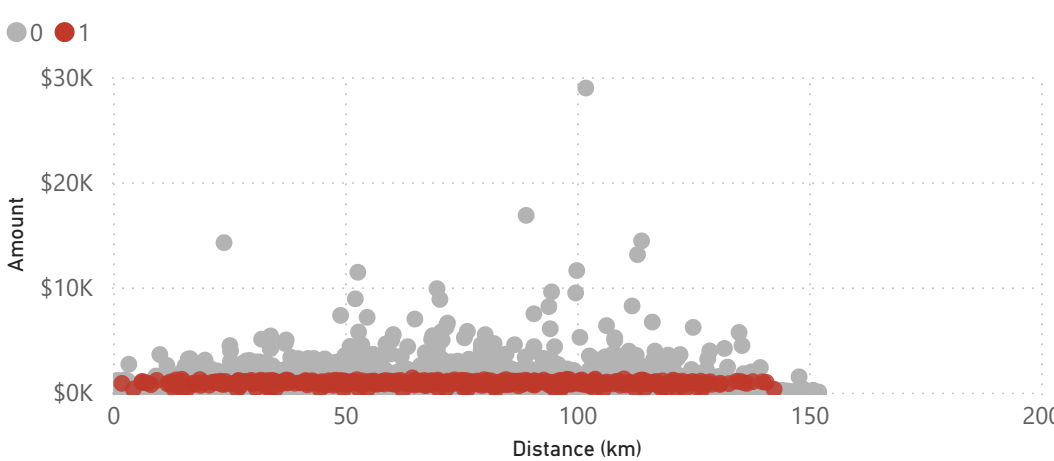
Fraud Rate by Amount Band



Total Fraud by Day of week



Transaction Amount Vs Distance



Customer Risk & Demographic Profile

Examines who is targeted, segmenting fraud by age, profession, and location to identify vulnerable groups and guide prevention.

Date Range

1/1/2019

12/31/2020

KEY INSIGHT

Fraud mostly **targeted at customers aged 18–29 (0.67%) and 75+ (0.75%)**, while middle-aged groups (30–44) face the **lowest exposure at 0.35%**. Alaska leads all states at a **1.69% fraud rate** - more than **3× the overall dataset average** - driven by a small but highly concentrated transaction pool.

- Executive Overview
- Deep Dive
- Risk Profile
- Model Performance
- Guide
- Clear all slicers

Age Group

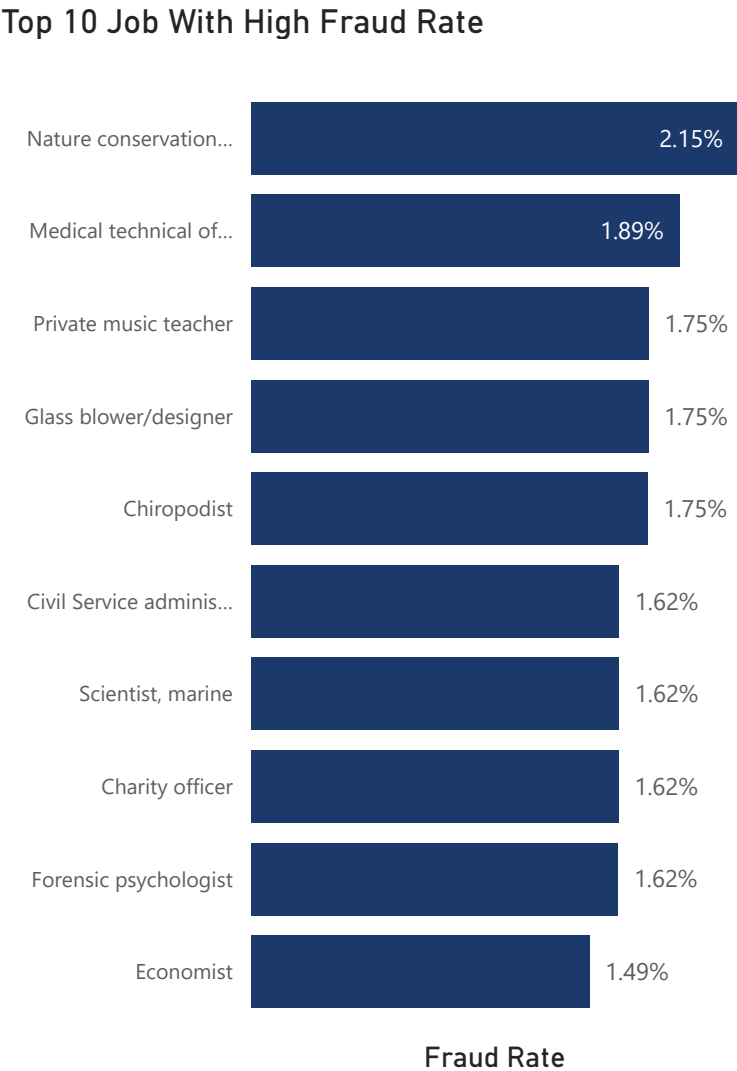
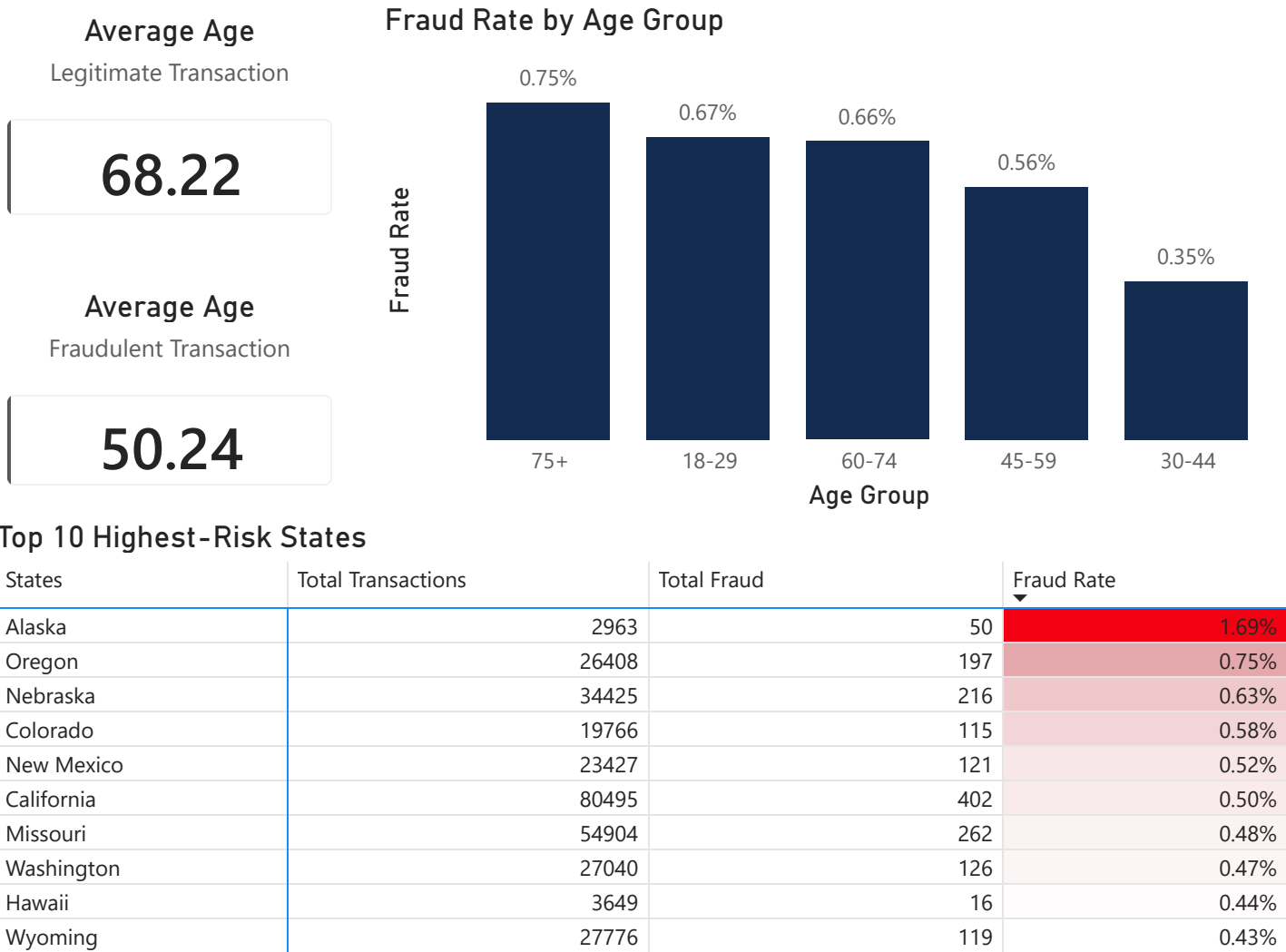
All

State

All

Amount Band

All



Model Performance & Accuracy Metrics

Evaluates fraud model performance using recall and precision to balance protection with minimizing false positives.

KEY INSIGHT

The model detects **75% of all fraud cases** (recall), deliberately prioritising capture over precision. The low precision of **2.32%** reflects a threshold tuned to minimise missed fraud - in a financial context, a missed case costs more than an unnecessary review. The **1,381 correctly caught fraud cases** represent an estimated **\$715,000 in prevented losses** (1,381 × \$518 average fraud amount).

